

Autonomous Stadium Litter Collection Robot

Sponsor: student self-sponsored

Background:

Game days are exciting but often generate large amounts of litter that are left behind in the stadium. After the game, maintenance crews spend several hours picking up and sorting through the trash and recycling discarded on the ground. This process is time-consuming, repetitive, and potentially dangerous to employees as they must climb up and down stairs, reach under bleachers, and potentially deal with broken glass.

An autonomous robot could assist maintenance crews by collecting litter and cleaning areas of the stadium, especially hard to reach areas. This will make their job faster, easier, and allow them to focus on more important and larger maintenance responsibilities.

Project Needs:

1. Robot should be able to autonomously and successfully navigate a stadium setting and avoid obstacles
2. Robot should be able to detect and effectively collect stadium litter items such as cans, cups, bottles, food containers, plastic, and wrappers
3. Solar rechargeable battery with a long enough battery life for robot to complete necessary tasks efficiently
4. Wireless communication system to monitor battery life, status, and allow emergency stop/control of the robot

Project Scope:

1. Design and build a robot that can maneuver around a stadium environment, including under seats
2. Design system to identify litter from surrounding obstacles
3. Design mechanism and container to pick up and store litter
4. Implement rechargeable solar battery energy storage system to power the robot
5. Develop communication system for robot status monitoring and safety
6. Test and document progress of robot navigation, detection, and communication systems

Potential Technical Considerations:

1. Sensors and cameras are needed to navigate the robot and identify obstacles from litter
2. Litter collection must be able to effectively store large quantities and a variety of litter
3. Battery and solar power systems must be efficient
4. Wireless communication system must be able to effectively and accurately update user on robot status, including a successful and safe emergency stop function

Deliverables:

1. Prototype robot that can autonomously navigate a stadium area
2. Functional trash collection container
3. Communication and object detection systems
4. Rechargeable battery and solar panel
5. Documentation of decisions made using the design and testing process

Assumptions and Dependencies: TBD